Linear Algebra (MTS203)

Midterm Exam, (120 mins — 20 Marks)

Spring 2024 Class no. 40241

Exam Date - March 11, 2024

IBA Institute of Business Administration Karachi
Leadership and Ideas for Tomorrow

Question - 1 (3 marks)

Find all solutions to the linear system:

$$\begin{aligned} x_1 - x_2 + 3x_3 + 2x_4 &= 1 \\ -x_1 + x_2 - 2x_3 + x_4 &= -2 \\ 2x_1 - 2x_2 + 7x_3 + 7x_4 &= 1 \end{aligned}$$

Question - 2 (2 marks)

Given:

$$A = \begin{bmatrix} \mathcal{I} & \mathcal{O} & \mathcal{C} \\ \mathcal{B} & \mathcal{I} & \mathcal{O} \\ \mathcal{O} & \mathcal{O} & \mathcal{I} \end{bmatrix}$$

where all of the submatrices are $n \times n$ blocks, determine the block form of A^{-1}.

Question - 3 (3 marks)

Let B and C be $n \times n$ matrices with the property that $B\boldsymbol{x} = C\boldsymbol{x}$ for all $\boldsymbol{x} \in \mathbb{R}^n$. Prove that $B = C$.

Question - 4 (3 marks)

Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ O & A_{22} \end{bmatrix}$$

where all four blocks are $n \times n$ matrices. If A_{11}, and A_{22} are nonsingular, show that A must also be nonsingular, and that A^{-1} must be of the form

$$\left[\begin{array}{c|c} A_{11}^{-1} & C \\ \hline O & A_{22}^{-1} \end{array}\right]$$

Determine C.

Question - 5 (3 marks)

What is an *involution*? Give its formula. Using this formula form 2 different matrices that are their own inverses.

Question - 6 (1 mark)

Suppose that a 3×3 matrix A is factored into a product:

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

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Determine $|A|$.

Question - ~~6~~ 7 (2 mark)
Show that if $\det(A) = 1$ then

$$\text{adj}(\text{adj } A) = A$$

Question - ~~7~~ 8 (2 mark)
Let R denote the set of real numbers. Define scalar multiplication by:

$$\alpha x = \alpha \cdot x \text{ (the usual multiplication of real numbers)}$$

and define vector addition $+$, by

$$x + y = \max(x, y) \text{ (the maximum of two numbers } x \text{ and } y\text{)}$$

Is R a vector space with these operations? Prove your answer.

Question - ~~8~~ (1 mark)
Is the set $\{1, x^2, x^2 - 2\}$ a spanning set for P_3? Prove your answer.

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QUESTION - 1 (3 marks)

Find all solutions to the linear system:

$$x_1 - x_2 + 3x_3 + 2x_4 = 1$$
$$-x_1 + x_2 - 2x_3 + x_4 = -2.$$
$$2x_1 - 2x_2 + 7x_3 + 7x_4 = 1$$

Sol.

$$\left[\begin{array}{cccc|c} 1 & -1 & 3 & 2 & 1 \\ -1 & 1 & -2 & 1 & -2 \\ 2 & -2 & 7 & 7 & 1 \end{array}\right]$$

$R_2 = R_2 + (1)R_1$ & $R_3 = R_3 + (-2)R_1$

$$\left[\begin{array}{cccc|c} 1 & -1 & 3 & 2 & 1 \\ 0 & 0 & 1 & 3 & -1 \\ 0 & 0 & 1 & 3 & -1 \end{array}\right]$$

$R_3 = R_3 + (-1)R_2$

$$\left[\begin{array}{cccc|c} 1 & -1 & 3 & 2 & 1 \\ 0 & 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$\Rightarrow \quad x_2 = \alpha.$

$x_4 = \beta.$

$x_3 + 3x_4 = -1. \quad \Rightarrow x_3 = -1 - 3\beta.$

and

$x_1 - x_2 + 3x_3 + 2x_4 = 1.$

$\Rightarrow \quad x_1 = 1 + \alpha + 3(-1-3\beta) + 2\beta$

$= 1 + \alpha + 3 + 9\beta + 2\beta.$

$x_1 = \alpha + 7\beta + 4$

$$\therefore \boxed{\bar{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \alpha + 7\beta + 4 \\ \alpha \\ -1-3\beta \\ \beta \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -1 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 7 \\ 0 \\ -3 \\ 1 \end{bmatrix}}$$ Ans

QUESTION - 2 (2 marks)

Given

$$A = \begin{bmatrix} I & 0 & C \\ B & I & 0 \\ 0 & 0 & I \end{bmatrix}$$

where all of the submatrices are $n \times n$ blocks, determine the block form of A^{-1}

$$\left[\begin{array}{ccc|ccc} I & 0 & C & I & 0 & 0 \\ B & I & 0 & 0 & I & 0 \\ 0 & 0 & I & 0 & 0 & I \end{array}\right]$$

$R_2 = R_2 + (-B) R_1$

$$\left[\begin{array}{ccc|ccc} I & 0 & C & I & 0 & 0 \\ 0 & I & -BC & -B & I & 0 \\ 0 & 0 & I & 0 & 0 & I \end{array}\right]$$

$R_1 = R_1 + (-C) R_3$

$$\left[\begin{array}{ccc|ccc} I & 0 & 0 & I & 0 & -C \\ 0 & I & -BC & -B & I & 0 \\ 0 & 0 & I & 0 & 0 & I \end{array}\right]$$

$R_2 = R_2 + (BC) R_3$.

$$\left[\begin{array}{ccc|ccc} I & 0 & 0 & I & 0 & -C \\ 0 & I & 0 & -B & I & BC \\ 0 & 0 & I & 0 & 0 & I \end{array}\right]$$

$$\therefore \boxed{A^{-1} = \begin{bmatrix} I & 0 & -C \\ -B & I & BC \\ 0 & 0 & I \end{bmatrix}}$$

Ans

QUESTION-3 (3 marks)

Let B and C be $n \times n$ matrices with the property that $B\bar{x} = C\bar{x}$ for all $\bar{x} \in \mathbb{R}^n$. Prove that $B = C$.

If $\quad B\bar{x} = C\bar{x} \qquad \forall\, x \in \mathbb{R}^n$

$\Rightarrow \quad B\bar{x} - C\bar{x} = \bar{0}$

$(B - C)\bar{x} = \bar{0} \qquad \forall\, x \in \mathbb{R}^n$

let

$A = B - C \qquad A = [\bar{a}_1\ \vec{a}_2, \cdots \vec{a}_n]$

$\Rightarrow \quad A\bar{x} = \bar{0} \qquad \forall\, x \in \mathbb{R}^n$

let $\bar{x} = e_j$ for $j = 1, 2, \cdots, n$.

then

$\bar{a}_j = A\bar{e}_j = 0 \qquad$ for $j = 1, 2, \cdots, n$.

and

hence A must be the zero matrix.

$\therefore \quad A = B - C = O \qquad$ ($\because$ O is $n \times n$ matrix whose all entries are zero)

$\Rightarrow \quad B - C = O$

$\boxed{B = C}$

proved

QUESTION - 4 (3 marks).

let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$$

where all far blocks are $n \times n$ matrices. If A_{11} & A_{22} are nonsingular show that A must be nonsingular, and that A^{-1} must be of the form.

$$\begin{bmatrix} A_{11}^{-1} & C \\ 0 & A_{22}^{-1} \end{bmatrix}$$

Determine C.

$$\begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} A_{11}^{-1} & C \\ 0 & A_{22}^{-1} \end{bmatrix} = \begin{bmatrix} A_{11} A_{11}^{-1} & A_{11} C + A_{12} A_{22}^{-1} \\ 0 & A_{22} A_{22}^{-1} \end{bmatrix}$$

$$= \begin{bmatrix} I & A_{11} C + A_{12} A_{22}^{-1} \\ 0 & I \end{bmatrix}$$

if $\quad A_{11} C + A_{12} A_{22}^{-1} = 0$

then

$$A_{11} C = - A_{12} A_{22}^{-1}$$

$$C = - A_{11}^{-1} A_{12} A_{22}^{-1}$$

let

$$\boxed{B = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1} A_{12} A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}}$$

$\therefore$

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1} A_{12} A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}$$

$$= \begin{bmatrix} A_{11} A_{11}^{-1} & -A_{11} A_{11}^{-1} A_{12} A_{22}^{-1} + A_{12} A_{22}^{-1} \\ 0 & A_{22} A_{22}^{-1} \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} = BA$$

$\Rightarrow \quad \boxed{B = A^{-1}}$

QUESTION - 5 (3 marks)

What is an involution? Give its formula. Using this formula form 2 different matrices that are their own inverses.

An $n \times n$ matrix 'A' is said to be an involution if $A^2 = I$

moreover $H = I - 2uu^T$ where $u \in \mathbb{R}^n$ is an arbitrary unit vector.

Let $u = (a_1, a_2, a_3, \ldots, a_n)^T$.

$$uu^T = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix} \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} = \begin{bmatrix} a_1^2 & a_1a_2 & a_1a_3 & \cdots & a_1a_n \\ a_2a_1 & a_2^2 & a_2a_3 & \cdots & a_2a_n \\ a_3a_1 & a_3a_2 & a_3^2 & \cdots & a_3a_n \\ \vdots & & & & \\ a_na_1 & \cdots & & & a_n^2 \end{bmatrix}$$

$$H = I - 2uu^T = \begin{bmatrix} 1-2a_1^2 & -2a_1a_2 & -2a_1a_3 & \cdots & -2a_1a_n \\ -2a_2a_1 & 1-2a_2^2 & -2a_2a_3 & \cdots & -2a_2a_n \\ \vdots & & \ddots & & \\ -2a_na_1 & \cdots & & & 1-2a_n^2 \end{bmatrix}$$

Mat 1

Now consider a concrete unit vector in $\mathbb{R}^3$.

$u = \left[\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right]^T$

Then $H = I - 2uu^T$ gives $\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$.

& $H^2 = I$

Mat 2

Let $u = \left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]^T \in \mathbb{R}^4$.

Then

$H = I - 2uu^T$ gives $\begin{bmatrix} 1/2 & -1/2 & -1/2 & -1/2 \\ -1/2 & 1/2 & -1/2 & -1/2 \\ -1/2 & -1/2 & 1/2 & -1/2 \\ -1/2 & -1/2 & -1/2 & 1/2 \end{bmatrix}$.

& $H^2 = I$.

QUESTION - 6 (1 mark)

Suppose that a 3x3 matrix A is factored into a product

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

determine $|A|$

$\because$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \quad \text{(Given)}$$

and let

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \quad \& \quad U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

then

$\det(A) = \det(LU)$

$\because \det(BC) = \det(B)\det(C)$

$\therefore \det(A) = \det(L)\det(U)$

and Since

"If M is an $n \times n$ triangular matrix, then the determinant of M equals the product of the diagonal elements of M".

$\therefore$

$$\det(A) = (1)(u_{11} \cdot u_{22} \cdot u_{33})$$

$$\Rightarrow \boxed{\det(A) = u_{11} u_{22} u_{33}} \text{ Ans}$$

✓

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QUESTION - 6 (2 marks)

Show that if $\det(A) = 1$ then

$\operatorname{adj}(\operatorname{adj} A) = A.$

$\because$ $\det(A) = 1.$

$\Rightarrow$ A is nonsingular.

and

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A.$$

$\Rightarrow$ $\operatorname{adj} A = \det(A) A^{-1}$

$\operatorname{adj} A = A^{-1}$ $\quad \because \det(A) = 1.$

and hence

$\operatorname{adj}(\operatorname{adj} A) = \operatorname{adj}(A^{-1}) \rightarrow (1)$

$\because$ $\operatorname{adj} A = \det(A) A^{-1}$

it follows that $\operatorname{adj} A$ is also nonsingular.

$$\Rightarrow (\operatorname{adj} A)^{-1} = \frac{1}{\det(A)} (A^{-1})^{-1}$$

$$= \det(A^{-1}) A. \rightarrow (2)$$

Also,

$$\operatorname{adj} A^{-1} = \det(A^{-1})(A^{-1})^{-1}$$

$\Rightarrow$ $\operatorname{adj} A^{-1} = \det(A^{-1}) A \rightarrow (3)$

$\therefore$ from (2) & (3) we have

$$\operatorname{adj} A^{-1} = (\operatorname{adj} A)^{-1} = \det(A^{-1}) A = \frac{1}{\det(A)} A$$

(1) $\Rightarrow$ $\operatorname{adj}(\operatorname{adj} A) = \frac{1}{\det A} A \Rightarrow \boxed{\operatorname{adj}(\operatorname{adj} A) = A}$ $\quad \because \det(A) = 1$

proved

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QUESTION 7 (2 marks)

Let $\mathbb{R}$ denote the set of real numbers. Define scalar multiplication by

$\alpha x = \alpha \cdot x$ (the usual multiplication of real numbers)

and define vector addition $+$, by

$x + y = \max(x, y)$ (the maximum of two numbers x & y)

Is $\mathbb{R}$ a vector space with these operations?

Prove your answer.

Axiom 1 : $\bar{x} + \bar{y} = \bar{y} + \bar{x}$ for any $\bar{x}$ & $\bar{y}$ in V

$\bar{x} + \bar{y} = \max(x, y)$ for any x & y in $\mathbb{R}$.

and $\bar{y} + \bar{x} = \max(y, x)$

$\Rightarrow \bar{x} + \bar{y} = \bar{y} + x$

Axiom 2 : $(\bar{x} + \bar{y}) + \bar{z} = \bar{x} + (\bar{y} + \bar{z})$ for any x, y & z in V

$(\bar{x} + \bar{y}) + \bar{z} = \max(x, y) + z$. for any x, y & z in $\mathbb{R}$

$= \max((x, y), z)$

$\bar{x} + (\bar{y} + \bar{z}) = x + \max(y, z)$

$= \max(x, (y, z))$

Axiom 3 : There exist an element $\bar{0}$ in V such that

$\bar{x} + \bar{0} = \bar{x}$ for each $\bar{x} \in V$.

$\bar{x} + \bar{0} = \max(x, 0)$ for each $\bar{x} \in \mathbb{R}$.

$\neq \bar{x}$ when $x < 0$

$\therefore$ Axiom 3 fail to hold

Hence

$\mathbb{R}$ is not a Vector Space.

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QUESTION 8 (1 mark)

Is the set $\{1, x^2, x^2-2\}$ a spanning set for P_3 ?
Prove your answer.

The vectors 1, x^2 and x^2-2 span P. Thus if $ax^2 + bx + c$ is any polynomial in P_3 it is possible to find scalars c_1, c_2, and c_3 such that

$$ax^2 + bx + c = c_1(1) + c_2(x^2) + c_3(x^2-2)$$

$$= c_1 + c_2x^2 + c_3x^2 - 2c_3.$$

$$= (c_2 + c_3)x^2 + 0x + (c_1 - 2c_3)$$

Setting

$$c_1 + c_3 = a$$

$$0 = b.$$

$$c_1 - 2c_3 = c.$$

✓

The above system is not consistent

Hence

the set $\{1, x^2, x^2-2\}$ is not a spanning set for P_3